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Assignment -05
Neural Networks
TCS-362

Q1. Numerical/Experimental: Analyze model accuracy improvements using transfer learning.

Ans. Transfer learning boosts accuracy by leveraging features learned from massive datasets (like ImageNet).

- Feature Extraction: Pre-trained layers act as advanced "detectors" for shapes and textures, reducing the risk of overfitting on small datasets.
- Fine-tuning: Adjusting the weights of top layers allows the model to adapt specifically to your data.
- Experimental Result: Typically, models

using transfer learning reach higher validation accuracy in significantly fewer epochs compared to training from scratch.

Q2. Case Study: Analyze deep learning-based face recognition system.

Ans. These systems generally follow a four-step pipeline:

- Detection: Finding faces in an image (e.g., using MTCNN).
- Alignment: Normalizing the face (eyes/nose position) for consistency.
- Feature Extraction: A CNN (like FaceNet) converts the image into a high-dimensional vector (embedding).
- Verification/Identification: Using Triplet Loss or Euclidean distance to compare embeddings. If the distance is small, it's a match.

Q3. Case Study: Explain NLP-based

sentiment analysis for social media data.

Ans. Social media data is "noisy" (slang, emojis, typos). The process involves:

- Preprocessing: Tokenization, removing handles (@), and converting emojis into text.
- Word Embeddings: Using Word2Vec or BERT to capture context and sarcasm.
- Classification: RNNs or LSTMs analyze the sequence of words to categorize the sentiment as Positive, Negative, or Neutral based on the emotional weight of the entire post.

Q4. Discuss advantages and limitations of deep learning tools such as Caffe, Theano, and Torch.

Ans. When comparing older deep learning frameworks, Caffe is widely recognized for its high speed and excellence in computer vision tasks due to its C++

core, though it remains notoriously difficult to customize and lacks flexibility for recurrent neural networks (RNNs). Theano was a pioneer in symbolic differentiation and optimized mathematical computations, but it suffered from high compile times and has since been largely deprecated. Meanwhile, Torch offered immense flexibility and fast prototyping capabilities through Lua; however, its reliance on a less popular language compared to Python and its steep learning curve eventually limited its widespread adoption.

Q5. Discuss sparse coding and representation learning.

Ans. - Sparse Coding: A technique where data is represented using a small number of "active" components from a large dictionary. It mimics how the human brain processes visual signals

efficiently, - Representation Learning:
Instead of manual feature engineering,
the model automatically learns the best
"features" (representations) from raw
data.

- The Link: Sparse coding is a form of
unsupervised representation learning that
helps models find the most essential
patterns while ignoring noise.